QUIRINO HIGH SCHOOL

TECHPRO • GRADE 11 ACTIVITY SHEET

CSS POSITIONING

Static, Relative, Absolute, Fixed, and Sticky

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Learning Objectives

- Explain the purpose of the CSS position property.
- Differentiate static, relative, absolute, fixed, and sticky positioning.
- Use top, right, bottom, and left to place elements.
- Create a web page demonstrating absolute, fixed, and sticky positioning.

Before You Begin

Use Visual Studio Code and a web browser. Create one index.html file. Save your work often, then refresh the browser to test every CSS change.

Starter Code

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>CSS Positioning Practice</title>
  <style>
    body { font-family: Arial, sans-serif; margin: 0; line-height: 1.6; padding-bottom: 80px; }
    header { background: #6b1024; color: white; padding: 20px; text-align: center; }
    main { padding: 20px; }
    .demo-box { padding: 12px; margin: 12px 0; border: 2px solid #6b1024; background: #f8e9ed; }
  </style>
</head>
<body>
  <header id="top">
    <h1>TechPro Positioning Lab</h1>
    <p>Explore how CSS places elements on a page.</p>
  </header>
  <main>
    <h2>Positioning Practice</h2>
    <div class="demo-box">Box 1: Normal flow</div>
    <div class="demo-box relative-box">Box 2: Relative positioning</div>
    <div class="demo-box">Box 3: Another normal-flow box</div>
    <p>Scroll down to test sticky and fixed elements.</p>
    <p>Practice paragraph 1. Observe how positioned elements behave.</p>
    <p>Practice paragraph 2. Keep experimenting with CSS values.</p>
    <p>Practice paragraph 3. Add more paragraphs to make the page scroll.</p>
  </main>
</body>
</html>
```

PART A — Understand the position Property

1. Create a folder named CSS_Positioning_Lab in Visual Studio Code. Create index.html and paste the starter code above.
2. Save the file and open it in a browser. Observe the three boxes before changing any CSS.
3. Add position: static; to .demo-box. Save and refresh the browser.
4. Static is the default. Static elements follow normal document flow and are not moved by top, right, bottom, or left.

PART B — Try position: relative

5. Add this CSS: .relative-box { position: relative; left: 30px; top: 10px; }
6. Save and refresh. Describe where Box 2 moved in relation to its original position.
7. Change left to 60px, then to -20px. Record what happens.
8. A relatively positioned element keeps its original space in normal document flow.

PART C — Create an Absolute Element

9. `Inside <main>, add: <div class="position-card"><p>Positioned container</p><span class="absolute-tag">NEW</span></div>`
10. `Add CSS: .position-card { position: relative; min-height: 120px; border: 2px solid #777; padding: 16px; margin: 20px 0; }`
11. `Add CSS: .absolute-tag { position: absolute; top: 8px; right: 8px; background: #6b1024; color: white; padding: 4px 8px; }`
12. Refresh. The NEW label should appear near the top-right of the card.
13. Temporarily remove position: relative from .position-card and observe the difference. Restore it afterward.
14. Absolute positioning uses the nearest positioned ancestor as its reference. It is removed from normal flow and may overlap other content.

PART D — Create a Fixed Help Button

15. Add this HTML just before </body>: Back to Top
16. Make sure the header has id="top".
17. `Add CSS: .help-button { position: fixed; right: 16px; bottom: 16px; background: #6b1024; color: white; padding: 10px 14px; text-decoration: none; border-radius: 6px; }`
18. Scroll the page. Observe that the button stays in the same place in the browser viewport.
19. Change bottom and right to 30px. Test again.
20. Fixed positioning uses the viewport as its reference and does not keep the element's original space in document flow.

PART E — Make a Sticky Heading

21. Add this heading inside <main>, before the practice paragraphs: <h2 class="sticky-heading">Sticky Section Heading</h2>
22. `Add CSS: .sticky-heading { position: sticky; top: 0; background: #f8e9ed; padding: 10px; border-bottom: 2px solid #6b1024; }`
23. Add several paragraphs so the page can scroll. Scroll slowly and watch the heading.
24. Try top: 10px and then top: 0. Compare the results.
25. Sticky behaves like relative until a scroll threshold is reached. At least one of top, right, bottom, or left must be set.

PART F — Compare and Experiment

26. Create a small labeled example for static, relative, absolute, fixed, and sticky.
27. Use top, right, bottom, or left to change the location of the positioned examples.
28. Write one sentence explaining the difference between absolute and fixed.
29. Write one sentence explaining the difference between fixed and sticky.
30. Test at a narrower browser width and make sure labels and buttons remain readable.

MINI-PROJECT — TechPro Learning Portal

Create a polished page that demonstrates the positioning methods you practiced.

- Build a one-page TechPro Learning Portal with a maroon header and a content card.
- Use a relative container with an absolutely positioned FEATURED label.
- Add a fixed Back to Top button in the lower-right corner.
- Add a sticky section heading with a top offset.
- Include at least five paragraphs or content blocks so scrolling can be tested.
- Keep text readable and prevent positioned elements from covering important content.

CHALLENGE TASK

- Add a notification badge positioned absolutely inside a relatively positioned profile card.
- Add a sticky navigation bar at the top of the page.
- Experiment with z-index if two positioned elements overlap. Record which element appears in front.

Check Your Work

☐ My HTML file opens without errors.

☐ I demonstrated static and relative positioning.

☐ My absolute element is inside a positioned container.

☐ My fixed button stays in place while scrolling.

☐ My sticky heading has a top, right, bottom, or left offset.

☐ I tested by scrolling and changing values.

☐ Content is readable and elements do not block important text.

Review Questions

1. What is the default value of the CSS position property?

   default value is static

2. What does position: relative do to an element?

   ffsets an element from its normal position while keeping its original space intact

3. Which value positions an element relative to its nearest positioned ancestor?

   absolute positioning

4. Which value keeps an element in the same place in the viewport while scrolling?

   fixed positioning

5. What must be specified for sticky positioning to work?

   an offset property must be set, such as top: 0

6. Name the four offset properties used with CSS positioning.

   top, right, bottom, and left

7. How is fixed positioning different from sticky positioning?

fixed positioning uses the viewport as a reference and stays in place during scrolling, whereas sticky positioning behaves like relative positioning until a scroll threshold is reached

8. Why is position: relative often added to a container holding an absolute label?

   it acts as the nearest positioned ancestor so that the absolute child element references it correctly

9. What happens to an absolutely positioned element's space in normal document flow?

   it is removed from the normal document flow

10. Give one example of when sticky positioning is useful.

a sticky section heading or navigation bar that scrolls with the page until it reaches a specific threshold (like top: 0) and locks into view

Quick Review

- static — default; follows normal document flow.
- relative — offset from normal position; original space remains.
- absolute — positioned relative to nearest positioned ancestor; removed from normal flow.
- fixed — positioned relative to viewport; stays in place during scrolling.
- sticky — behaves like relative until a scroll threshold; needs an offset such as top: 0.
- top, right, bottom, and left set an element's offset.

References

- https://www.w3schools.com/css/css_position.asp
- https://www.w3schools.com/css/css_positioning_fixed_absolute.asp
- https://www.w3schools.com/css/css_positioning_sticky.asp